

# After Attention: The Economics of Markets Where the Buyer Is an Agent

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## Abstract

The consumer economy runs in part on *attention rent*: margin that sellers extract because human buyers search a little, remember imperfectly, and respond to persuasion. As personal purchasing agents ("twins") take over consumer choice, the standard expectation is that this rent collapses. We model a market with a mixed population of human and agent buyers in which sellers choose price, consumer-directed advertising, and agent-directed optimization spend, and we show that the collapse is neither smooth nor complete. Three results follow from the structure of agent demand. First, because a well-informed agent's demand is close to winner-take-most in quality-adjusted price, sellers face a discrete choice between a human-exploiting price and an agent-competitive one, which produces a *tipping point*: in our simulations gross margin falls 1.4 percentage points as the agent share rises from 0 to 0.2, then drops from 40% to 33% by a share of 0.3 and to 4% when all buyers are agents. Second, the mixed population admits no pure-strategy equilibrium in the interior — a discrete-grid counterpart of Varian (1980) — which we prove and which matches the numerics exactly: every convergent run in our sweeps sits at one of the two boundaries. Third, if agents are persuadable, advertising does not disappear but *migrates*: total persuasion spending peaks at 1.68 times the pre-agent advertising budget and remains above it when all buyers are agents. That migration does not chiefly misallocate demand — good sellers still win about 85% of sales either way — but it preserves seller rent (margin 9.9% against 4.0%), cuts consumer net value per unit by about 80%, and concentrates the market by a factor of 2.4 (HHI 7,424 against 3,102). We also read the agentic-commerce protocols standardized between 2025 and 2026 — AP2, ACP, Visa's Trusted Agent Protocol, Mastercard Agent Pay and UCP — and find that they cryptographically secure payment authorization while leaving product claims unstructured and unattested, lowering the friction that governs adoption without touching the parameter that governs who captures the gains. Simulation code and all runs are public.

**Keywords:** agentic commerce, attention rent, buyer agents, persuasion, market efficiency, AI agents **JEL:** D83, L13, M37, D47

## 1. Introduction

Roughly \$1.3 trillion of global advertising revenue is forecast for 2026, excluding US political advertising (WPP Media 2026). A large share of that spending exists because human buyers do not compare exhaustively: they consider a handful of options, weight familiarity, and can be moved by presentation. Following Stigler (1961), we call the resulting margin *attention rent* — the part of a seller's markup that survives only because buyers' search and recall are limited and their preferences are manipulable.

Purchasing agents attack that rent directly. An agent can enumerate a catalog, hold prices and attributes in memory, and choose without fatigue. Forecasts of adoption are substantial: Morgan Stanley (2025) projects \$190–385 billion of US e-commerce spending by agentic shoppers by 2030, and McKinsey (2025) puts "orchestrated revenue" in US B2C retail at \$900 billion to \$1 trillion by 2030, with a global range of \$3–5 trillion. The payment rails are already shipping — Mastercard Agent Pay (April 2025), Google's Agent

Payments Protocol (September 2025), OpenAI and Stripe's Instant Checkout and Agentic Commerce Protocol (September 2025), Visa's Trusted Agent Protocol (October 2025), and the Universal Commerce Protocol co-developed by Google with Shopify, Etsy, Wayfair, Target and Walmart (January 2026). Early displacement of human attention is already measurable in adjacent behavior: Pew Research Center (2025) found that users clicked a traditional search result on 8% of visits to Google searches returning an AI summary, versus 15% of visits without one.

The natural inference is that attention rent collapses and the advertising that harvests it goes with it. We argue that this inference is right in direction and wrong in shape, for two reasons that both follow from what agent demand looks like from a seller's side.

**The first is discreteness.** A well-informed agent does not spread its purchases across a consideration set in proportion to persuasion; it selects the best quality-adjusted price it can find. Seller demand from the agent segment is therefore close to winner-take-most. A seller choosing a price faces two distinct candidate optima — a high price that harvests the human segment and forgoes the agent segment, or a low price that competes for both — and switches between them discontinuously as the agent share crosses a threshold. Margins hold up, then fall off a cliff. Brands are safe right up until they are not.

**The second is substitutability of persuasion targets.** Advertising is not a terminal expense category; it is the current best instrument for moving buyer choice. If an agent's choice can be shifted by how a listing is written, structured, or ranked, then "agent-directed optimization" is simply the instrument that replaces advertising, and sellers will fund it for the same reason they funded advertising. Whether total persuasion spending falls or rises then depends on the marginal return to persuading an agent relative to persuading a person. Because agent demand is concentrated — winning an agent's choice wins all of its purchases, not a share of its attention — that marginal return can be *higher*, which turns the transition into an arms race rather than a wind-down.

Both mechanisms have empirical footing. Allouah et al. (2026) show that LLM buyers exhibit position bias, concentrate demand on modal products, and shift market share in response to query-conditional description edits. Salvi et al. (2026) find in two preregistered experiments ( $n = 2,012$ ) that sponsored products were selected 61.2% of the time in AI-mediated conversations versus 22.4% in search, with detection below 10% when commercial intent was concealed. Kumar and Lakkaraju (2024) demonstrate that inserting a strategic text sequence into product information raises an LLM's recommendation probability. Persuadable agents are not a hypothetical.

**Contribution.** The paper makes four claims. The first is a share-dependent tipping point in attention rent: we give the condition under which sellers abandon the human-exploiting price (Proposition 1) and show that the switch is discontinuous in the agent share rather than gradual. The second is a migration condition for persuasion spending: we give the condition under which sellers fund agent-directed optimization (Proposition 2), and show numerically that total persuasion spending peaks above the pre-agent advertising budget. A third formal result (Proposition 3) establishes that the mixed population admits no pure-strategy equilibrium in the interior, which explains the cycling that a best-response dynamic exhibits there and is confirmed exactly by the numerics. The third is a reading of the deployed rails: we compare the published agentic-commerce specifications against what an evidence-consuming agent would need, and find that they standardize authorization rather than attestation (§3). The fourth is the resulting claim about distribution: in our model the single parameter with the largest effect on who captures the gains from agent-mediated commerce — seller margin, consumer surplus and market concentration — is not the number of sellers or the cost of search but the agents' resistance to persuasion, which makes an agent's duty of loyalty to its principal an economic mechanism rather than an ethical garnish.

We also name what we do not claim. We do not estimate the persuadability of any deployed agent; our parameter for it is stylized, and calibrating it is the most valuable next step (§10). We do not model platform-owned agents with divided loyalties, though we expect that case to strengthen P3. And our numerical results come from a best-response dynamic that reaches a pure-strategy equilibrium in only 12 of 110 runs; we report averages over cycling behavior and treat magnitudes as illustrative and directions as the finding.

All proofs are in Appendix A; simulation implementation and robustness are in Appendices B and C.

This paper is a sequel to *The Autonomous Agentic Store* (Vissa 2026), which automated the seller side; here we automate the buyer and ask what the market becomes.

## 2. Related work

**Agent-buyer economics.** The closest theoretical work is Lucier et al. (2026), which studies the equilibrium welfare effects of improving consumer search and finds, among other results, that more informative search can reduce consumer surplus. Our focus differs in the object of study: we model persuasion spending as a choice variable on the seller side and make the *composition* of the buyer population the driver, which is what produces a threshold rather than a comparative-statics result. Dong, Luo and Xu (2026) formalize agentic versus manual search around preference articulation and catalog breadth; they have no advertising, no brands, and no adversarial sellers. Shahidi et al. (2025) frame agents as a collapse in transaction costs with implications for market design, but do not model attention rent or its migration. Turner-Smith et al. (2026) design truthful pricing for advertisements placed inside AI-generated responses — the mechanism-design counterpart to our P2, though they take agent mediation as given rather than deriving the migration of spend from a rising agent share.

**Evidence on agent manipulability.** Allouah et al. (2026), Kumar and Lakkaraju (2024), and Salvi et al. (2026) supply the empirical basis for a positive persuadability parameter. Wadi and Ma (2026) complicate the story usefully: across 5,000 LLM shopping sessions on hotel listings they find position predicts inspection only weakly and non-monotonically, which is evidence that *placement* rent in particular erodes even where persuadability persists. Aggarwal et al. (2024) document generative engine optimization as a practice, i.e. the supply side of agent-directed persuasion already exists commercially. Affonso (2026) models vertical tacit collusion in AI-mediated markets, calibrating compounding platform-ranking and seller-description exploitation, and finds joint harm more than double the independent case; that paper is the nearest existing work to our P3, and our contribution relative to it is to place robustness on a continuum and trace market outcomes across the adoption path.

**Informed and uninformed buyers.** The closest classical antecedent is the literature on markets containing both shoppers who compare and shoppers who do not. Salop and Stiglitz (1977) show that a population split between informed and uninformed buyers sustains price dispersion rather than convergence; Varian (1980) builds the canonical model in which sellers mix between a high price aimed at captive buyers and a low price aimed at shoppers, so that no pure-strategy equilibrium exists; Burdett and Judd (1983) derive dispersion from noisy search alone. A purchasing agent is, in this tradition, an informed buyer whose share of the population is rising. Our contribution relative to that literature is to make the *share itself* the comparative static of interest, to let sellers choose how much to spend persuading each type rather than only what price to post, and to allow the informed buyer to be manipulable — which is the case those models do not contain, because a shopper who compares perfectly cannot be persuaded.

**Search, persuasion and brand economics.** Our framing inherits from Stigler (1961) on the economics of information, Diamond (1971) on the fragility of competitive pricing under search frictions, Bakos (1997) on

buyer search costs in electronic marketplaces, and Ellison and Ellison (2009) on obfuscation as a response to falling search costs. The central empirical analogue is Bronnenberg et al. (2015): pharmacists and physicians — informed shoppers — pay substantially smaller brand premiums than other buyers. Purchasing agents are, in effect, informed shoppers at scale, which is why a large share of the brand premium should be expected to be contestable. Gal and Elkin-Koren (2017) anticipated the legal shape of algorithmic consumers. Bansal et al. (2025) provide an open simulation environment for agentic markets that is complementary to the model here.

**Positioning.** The distinct claims of this paper are (i) the share-dependent discontinuity in attention rent, (ii) persuasion-spend migration with a possible increase in total persuasion spending, and (iii) manipulation resistance as the dominant efficiency parameter. Where prior work overlaps a claim, it does so on a different margin: Wadi and Ma on placement rents empirically, Turner-Smith et al. on pricing agent-directed ads, Lucier et al. on search informativeness in equilibrium, Affonso on compounding exploitation.

### 3. What the deployed rails standardize — and what they leave open

The economics above are not being written on a blank surface. Between April 2025 and January 2026 the payment and commerce industry standardized five protocols for agent-mediated purchasing, and their contents matter for this paper's argument, because they determine which of the two regimes in our model the market is being built toward. We read the current published specifications — Mastercard Agent Pay (April 2025), Google's Agent Payments Protocol AP2 (v0.1 September 2025, v0.2 current), OpenAI and Stripe's Agentic Commerce Protocol and Instant Checkout (September 2025), Visa's Trusted Agent Protocol (October 2025), and the Universal Commerce Protocol UCP (v2026-01-11, v2026-08-25 current) — with one question in mind: do they give a purchasing agent any means of distinguishing an attested fact from a persuasive claim?

They do not. What they standardize with real cryptographic care is *authorization*, and what they leave unstructured is *evidence*.

**Signing is applied to the envelope, never to the fact.** UCP's baseline requires verifying ES256 signatures over the HTTP message — method, authority, path and headers, with the body bound by a SHA-256 digest. AP2's merchant signs the entire Checkout payload as one JWT covering line items, total price and policies, and that signature is nested inside a user- or agent-signed mandate. In neither protocol is there an object of the form {product identifier, attribute, value, issuer identifier, signature}. A seller's assertion that a garment is organic cotton, or that a component ships from a particular country, travels as ordinary text inside a signed envelope. The signature proves the merchant sent the message; it says nothing about whether the claim is true, and there is no independent issuer whose attestation could say so.

**Product descriptions are one undifferentiated layer.** UCP's product and variant objects carry title, description, media, tags, rating and a free-form metadata field described as "business-defined custom data." Nothing marks which of that is verifiable and which is persuasion. AP2's item object carries only an identifier and a title, so the question does not arise there at all. An agent consuming either format has no structural basis for weighting the claim "ships in two days" differently from the claim "customers love it."

**Trust is bilateral and payment-scoped.** UCP resolves keys from self-published JWK sets at a well-known endpoint; AP2 offers a user-credential model and a trusted-agent-provider model, and lists public key issuance and distribution among its open problems. Neither defines a registry that answers the question an evidence-consuming agent actually needs answered: *is this issuer authoritative for this attribute type?* Trust in these protocols establishes who may authorize a payment, not who may certify a fact.

**The catalog layer disclaims itself.** UCP is explicit that catalog responses "are not transactional commitments — checkout is authoritative," and that responses "SHOULD NOT be reused across sessions without re-validation." This is a reasonable engineering decision about freshness. It is also a statement that the layer where product claims live is the layer the standard declines to bind.

**Agent reasoning is out of scope by design.** Neither specification contains any notion of confidence, verification level, or weighting of unverified content. AP2's stated pillar is verifiable intent — "deterministic, non-repudiable proof of intent from all parties" — which is binary authorization proof about the buyer's wishes, not a graded assessment of the seller's claims. Both protocols stop precisely where the buyer agent's judgment begins.

**Why this matters for P1–P3.** In the language of our model, these protocols are a large and well-engineered reduction in the friction of agent purchasing, which raises the agent share  $\alpha$ . They contain nothing that acts on  $\gamma$ . The infrastructure being deployed accelerates the transition along the horizontal axis of every panel in Figure 1 while leaving the parameter that selects between the two curves entirely to the agent's builder. If our P2 is right, that is the configuration in which persuasion spending migrates rather than disappears: adoption rises quickly, and nothing in the rails makes a seller's attested claim cheaper to transmit than an unattested one.

This also sharpens what an intervention would have to look like. A registry of agents, such as the one contemplated in the AI AGENT Act of 2026, addresses identity and access. A protocol that signs the checkout addresses authorization. Neither addresses whether a seller's claims about a product can be checked by the agent that reads them, which is the quantity  $\gamma$  measures. We return to this in §8.

One caveat on method. These are living specifications: UCP has issued four versions in eight months, and both protocols have extension mechanisms. We report what the specifications published as of September 2026 require and permit. An implementation may do more than its specification demands, and a future version may add attribute-level attestation — which would be, on our reading, the single most consequential change the standards could make.

## 4. Model

A single product category has  $N$  sellers and  $M$  buyers. Seller  $j$  has true quality  $q_j \sim U(0,1)$  and marginal cost  $c_j = 5 + 10 q_j$ , so higher quality is genuinely more expensive to produce. Each seller chooses a markup  $m_j$  (price  $p_j = c_j(1+m_j)$ ), consumer-directed advertising  $A_j$ , and agent-directed optimization spend  $G_j$ .

A share  $\alpha$  of buyers are **twins** (purchasing agents acting for a principal); the remaining share  $(1-\alpha)$  are **humans**. Both types share the same underlying tastes; they differ in what they see and what moves them.

**Humans.** A human considers seller  $j$  with probability  $\pi(A_j) = \sigma(a_0 + a_1 \log(1+A_j))$ : advertising buys entry into the consideration set. Conditional on consideration, utility is

$$U^h_{ij} = V_0 + \theta_q(q_j + \varepsilon_{ij}) - \lambda p_j + \beta \log(1+A_j) + \eta_{ij},$$

where  $\varepsilon_{ij}$  is a perception error on quality,  $\beta$  is the persuasion weight (advertising shifts perceived value, not just awareness), and  $\eta_{ij}$  is an i.i.d. extreme-value taste shock. Humans thus have limited consideration, noisy quality perception, and persuadable valuations — the three ingredients of attention rent.

**Twins.** A twin considers every seller, perceives quality without error, and chooses more deterministically:

$$U^t_{ij} = \mu_t(V_0 + \theta_q q_j - \lambda p_j + \gamma \log(1+G_j)) + \eta_{ij},$$

with  $\mu_t > 1$  sharpening choice. The parameter  $\gamma \geq 0$  is the twin's **persuadability**: the weight it places on agent-directed optimization that carries no true value.  $\gamma = 0$  describes a robust twin that cannot be moved by presentation;  $\gamma > 0$  describes one that can. Both types may decline to purchase (an outside option drawn from the same extreme-value family).

**Sellers.** Sellers maximize expected profit  $\Pi_j = (p_j - c_j)D_j(m, A, G) - A_j - G_j$  over a discrete grid of markups, advertising levels and agent-spend levels, by iterated best response in randomized order, holding rivals fixed, with common random numbers across candidate actions for variance reduction.

Two features of this specification do the work in what follows. First, twins' sharper choice rule makes their demand close to winner-take-most in quality-adjusted price: a small price advantage moves a large block of twin demand. Second,  $A$  and  $G$  enter separately and target disjoint segments, so the model can express substitution between persuading people and persuading agents rather than assuming persuasion disappears.

## 5. Analytical results

This section states the two propositions formally. Both are partial-equilibrium results: they characterize one seller's best response with rivals' actions held fixed, which is the step the simulation in §6–7 iterates to convergence or to a cycle. **All proofs are in Appendix A.**

We work under three assumptions.

**Assumption 1 (Two-candidate pricing).** Prices are chosen from a finite grid. For seller  $j$ , let  $p_H^j$  be the price maximizing profit from the human segment alone, and let  $p_T^j$  be the highest price at which  $j$  wins twin demand given rivals' prices, with  $p_T^j < p_H^j$ .

**Assumption 2 (Winner-take-most twin demand).** As the twin logit scale  $\mu_t$  grows, twin demand concentrates on the seller maximizing quality-adjusted surplus  $\theta_q q_j - \lambda p_j + \gamma \log(1+G_j)$ : seller  $j$  receives the twin segment if it is the maximizer and approximately none of it otherwise.

**Assumption 3 (Fixed rivals).** Rivals' prices and spends are held at their current values while  $j$  best-responds.

### 5.1 The tipping point (Proposition 1)

Let  $\Pi_H$  be per-human profit at  $p_H^j$ , and let  $\pi^h_T$  and  $\pi^t_T$  be per-human and per-twin profit at  $p_T^j$ , with  $\Delta \equiv \Pi_H - \pi^h_T > 0$  and  $\pi^t_T > 0$ .

**Proposition 1.** Under Assumptions 1–3, seller  $j$  prefers the human-exploiting price  $p_H^j$  if and only if the twin share satisfies  $\alpha \leq \alpha^*_j$ , where

$$\alpha^*_j = (\Delta) / (\Delta + \pi^t_T) = (\Pi_H - \pi^h_T) / (\Pi_H - \pi^h_T + \pi^t_T) \in (0,1).$$

Moreover  $\alpha^*_j$  is strictly decreasing in  $\pi^t_T$  and strictly increasing in  $\Delta$ .

**Corollary 1 (Discontinuity).** The seller's optimal price is a step function of  $\alpha$  with a single jump at  $\alpha^*_j$ . *Gross margin therefore does not decline smoothly in the twin share: it is locally flat, then falls discontinuously. Across heterogeneous sellers the aggregate margin path is a weighted sum of such steps, which smooths the jump into a knee located where the distribution of  $\alpha^*_j$  has mass.*

Two readings follow. Categories where twins buy in volume — high  $\pi^t_T$  — tip at lower adoption. And a seller observing stable margins during early agent adoption learns nothing reassuring, because stability is what Proposition 1 predicts right up to  $\alpha^*_j$ .

## 5.2 Migration of persuasion spending (Proposition 2)

Write  $s^t_j$  for seller  $j$ 's share of the twin segment and  $s^h_j$  for its share of the human segment. The seller funds each instrument up to the point where marginal revenue equals its unit cost.

**Proposition 2.** Under Assumptions 2–3, with twin choice probabilities given by the logit form of §4, seller  $j$  funds strictly positive agent-directed spend if and only if

$$\gamma > \gamma^*_j \equiv (1)/(\alpha M (p_j - c_j) \mu_t s^t_j (1 - s^t_j)).$$

The threshold  $\gamma^*_j$  is strictly decreasing in the twin share  $\alpha$ , in the margin  $p_j - c_j$ , and in the choice sharpness  $\mu_t$ .

The comparative static on  $\mu_t$  is the substantive one. The same parameter that makes twin demand winner-take-most — and so drives the margin collapse of Proposition 1 — also raises the marginal return to persuading a twin, because winning a twin's choice wins its entire demand rather than a share of its attention. Sharper agents are both more efficient buyers when they cannot be persuaded and more valuable targets when they can.

Two boundary cases. At  $\gamma = 0$  the condition fails for every  $\alpha$ , so  $G_j = 0$  identically and advertising declines toward zero as  $\alpha \rightarrow 1$  because the segment it addresses disappears. At large  $\gamma$  the condition binds at modest  $\alpha$ , and agent-directed spending begins well before twins are a majority of buyers.

**What Proposition 2 does not establish.** The funding condition says when agent-directed spend is positive, not how large total persuasion spending  $\$A + \$G$  becomes. Whether the total exceeds its pre-agent level depends on equilibrium interaction among sellers and is a numerical result, reported in §7.2, not a theorem.

## 5.3 No pure-strategy equilibrium in the interior (Proposition 3)

The mixed population creates a tension familiar from Varian (1980): a seller wants a high price to harvest buyers who do not compare, and a low price to win buyers who do. On a discrete price grid this tension has a sharp consequence.

**Assumption 4 (Duopoly benchmark).** Two symmetric sellers with common marginal cost  $c$ . Each has captive human demand  $H/2$  with reservation price  $v_H$ ; humans do not compare. Twin demand  $T$  goes entirely to the strictly lower-priced seller, split evenly at a tie. Prices lie on the grid  $c + k\delta$  with maximum  $v_H$ .

**Proposition 3.** Under Assumption 4, if

$$(i) (v_H - c) T > 2\delta((H)/(2) + T) \text{ and } (ii) (v_H - c)(H)/(2) > (2\delta(H)/(2) + T)((H)/(2) + (T)/(2))T,$$

then the pricing game has no pure-strategy Nash equilibrium.

**Corollary 2.** Writing  $T = \alpha M$  and  $H = (1-\alpha)M$ , conditions (i) and (ii) hold simultaneously for an interval of interior twin shares, and as the grid is refined ( $\delta \rightarrow 0$ ) that interval expands to all of  $(0,1)$ . At the boundaries the tension vanishes: at  $\alpha = 0$  there is no segment to undercut for, and at  $\alpha = 1$  there are no captive buyers to harvest.

This is the discrete-grid counterpart of the classical non-existence results, and it makes the cycling reported in §6 a prediction rather than a defect. Equilibrium is recovered in mixed strategies — the setting of Dasgupta and Maskin (1986) for games with discontinuous payoffs — which is what the smoothed best-response dynamic of §6 approximates numerically.

## 5.4 Integrity as the efficiency parameter

Propositions 1 and 2 together locate the efficiency of an agent-mediated market in a single parameter. At  $\gamma = 0$ , Assumption 2 implies the twin segment allocates to the seller maximizing true quality-adjusted surplus, and Proposition 1 drives margins toward cost as  $\alpha$  rises. At  $\gamma > 0$ , the same concentration of demand operates on  $\theta_{-q} q_{-j} - \lambda p_{-j} + \gamma \log(1+G_{-j})$ , so the sharper the agent, the more completely demand follows whoever spends most on being chosen. Neither the number of sellers nor the cost of search appears in this comparison. That is the formal content of the claim that sharpening a buyer agent without securing its loyalty relocates the rent rather than dissolving it — quantified in §7.3.

## 6. Simulation

We instantiate the model with  $N = 12$  sellers and  $M = 3\{, \}000$  buyers;  $\theta_{-q} = 12$ ,  $\lambda = 1$ ,  $V_{-0} = 5$ , human persuasion weight  $\beta = 1.2$ , human quality-perception noise  $\sigma\epsilon = 1$ , twin logit scale  $\mu_{-t} = 3$ , consideration parameters  $a_{-0} = -1.5$ ,  $a_{-1} = 0.9$ . Markups are drawn from a 16-point grid spanning 1% to 170%;  $A$  and  $G$  from an 8-point grid spanning 0 to 1,500.

**Solution concept.** Proposition 3 says the interior of this game has no pure-strategy equilibrium, so iterating exact best responses produces cycles rather than convergence, and the resulting numbers depend on where the cycle is cut. We therefore use *smoothed* best response: a revising seller draws its action from a logit distribution over candidate profits with temperature  $\tau$  proportional to the dispersion of those profits, and revises with probability  $1 - \iota$  in any given round ( $\tau = 0.15$ ,  $\iota = 0.35$ ). This is the standard perturbation used to select among mixed equilibria in games of this class and it yields a stationary distribution to average over rather than a cycle to sample arbitrarily. Outcomes are computed at each of the final 10 of 40 iterations and averaged, then averaged over **20 seeds**.

We sweep the agent share  $\alpha \in 0, 0.1, \dots, 1.0$  under two persuadability regimes — **robust twins** ( $\gamma = 0$ ) and **manipulable twins** ( $\gamma = 1.5$ ) — giving 440 runs. Appendix B documents the implementation; Appendix C reports what the smoothing costs and where the numbers are fragile.

**A correction to an earlier draft.** An earlier version of this simulation used a coarser action grid (12 markup points) and exact best response. On that configuration the fully manipulable market appeared to route 95% of sales away from the best-value sellers. That result does not survive grid refinement — under either dynamic — and we report it here because it is the kind of artifact a coarse discrete action space produces: with few price points, a single seller that outspends on agent-directed optimization can capture the whole twin segment, and nothing in the grid allows rivals a fine enough response. The findings below are those that survive.

## 7. Results

Table 1 reports outcomes at five values of the agent share under both regimes; all figures are means over 20 seeds. Figure 1 plots the full sweep.

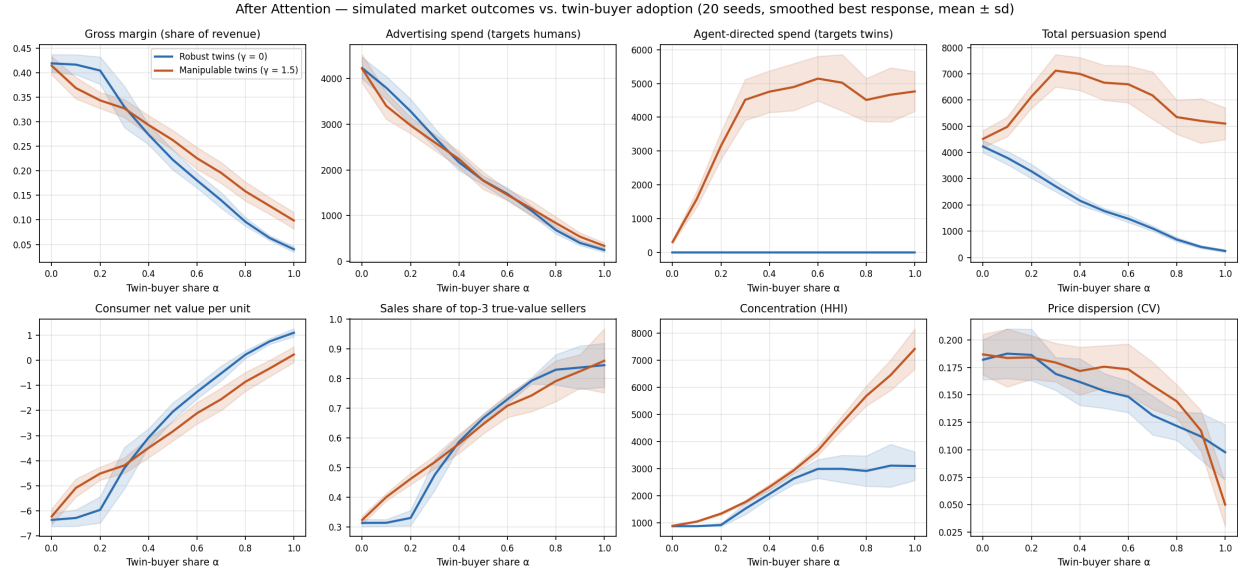


Figure 1. Simulated market outcomes against twin-buyer share  $\alpha$ , for robust ( $\gamma = 0$ ) and manipulable ( $\gamma = 1.5$ ) twins; 20 seeds, smoothed best response, mean  $\pm$  sd. Panels: gross margin, advertising spend, agent-directed spend, total persuasion spend, consumer net value per unit, sales share of top-3 true-value sellers, concentration (HHI), price dispersion.

Table 1. Market outcomes vs. agent-buyer share (20 seeds, smoothed best response)

| Outcome                                 | Regime      | $\alpha = 0$ | $\alpha = 0.2$ | $\alpha = 0.3$ | $\alpha = 0.5$ | $\alpha = 1$ |
|-----------------------------------------|-------------|--------------|----------------|----------------|----------------|--------------|
| Gross margin (share of revenue)         | robust      | 41.9%        | 40.5%          | 33.1%          | 22.3%          | <b>4.0%</b>  |
|                                         | manipulable | 41.4%        | 34.4%          | 32.7%          | 26.3%          | <b>9.9%</b>  |
| Advertising spend                       | robust      | 4,228        | 3,283          | 2,709          | 1,770          | 247          |
|                                         | manipulable | 4,214        | 2,975          | 2,601          | 1,765          | 337          |
| Agent-directed spend                    | robust      | 0            | 0              | 0              | 0              | 0            |
|                                         | manipulable | 304          | 3,155          | 4,518          | 4,896          | 4,765        |
| Total persuasion spend                  | robust      | 4,228        | 3,283          | 2,709          | 1,770          | 247          |
|                                         | manipulable | 4,518        | 6,130          | <b>7,119</b>   | 6,661          | 5,102        |
| Sales share of top-3 true-value sellers | robust      | 31.4%        | 33.1%          | 47.6%          | 66.6%          | 84.5%        |
|                                         | manipulable | 32.4%        | 46.2%          | 51.9%          | 64.6%          | 85.9%        |
| Consumer net value per unit             | robust      | -6.36        | -5.96          | -4.30          | -2.05          | <b>+1.09</b> |
|                                         | manipulable | -6.22        | -4.51          | -4.19          | -2.84          | <b>+0.23</b> |
| Concentration (HHI)                     | robust      | 883          | 927            | 1,527          | 2,640          | 3,102        |
|                                         | manipulable | 897          | 1,346          | 1,773          | 2,941          | <b>7,424</b> |
| Price dispersion (CV)                   | robust      | 0.182        | 0.186          | 0.169          | 0.154          | 0.098        |
|                                         | manipulable | 0.187        | 0.184          | 0.179          | 0.176          | 0.050        |

## 7.1 Attention rent collapses with a knee, not a slope (P1)

With robust agents, gross margin is 41.9% with no agents, 41.7% at a 10% agent share and 40.5% at 20% — a move of 1.4 percentage points across a fifth of the market. It then falls to 33.1% at  $\alpha = 0.3$ , 22.3% at  $\alpha = 0.5$ , and 4.0% when every buyer is an agent. The knee between  $\alpha = 0.2$  and  $\alpha = 0.3$  is the aggregate

signature of the switch in Proposition 1: below it, sellers hold the human-exploiting price and treat lost agent demand as an acceptable cost; above it, the agent segment is too large to forgo.

Advertising follows the segment it addresses, falling from 4,228 to 247 as  $\alpha$  goes from 0 to 1. The residual at  $\alpha = 1$  is smoothing noise rather than optimal spending — with no humans left to reach, advertising has no return, and Appendix C uses this residual as the noise floor for reading the other spend series.

Allocation improves substantially: the sellers offering the best true quality-adjusted value take 31.4% of sales in the all-human market and 84.5% in the all-agent market, consumer net value per unit rises from  $-6.36$  to  $+1.09$ , and price dispersion falls from a CV of 0.182 to 0.098. Robust agents deliver roughly what the efficiency argument for them promises.

The practical implication of the knee is a forecasting one. A brand tracking margin against agent adoption sees almost nothing until adoption approaches a quarter of its buyers, at which point the mechanism supporting its price gives way over a small further increment. Flat margins during early agent adoption are not evidence of resilience.

## 7.2 Persuasion migrates and exceeds the pre-agent budget (P2)

With persuadable agents, agent-directed spend appears as soon as there are agents to address — 3,155 at  $\alpha = 0.2$ , rising to about 4,800 and staying there — while advertising declines along essentially the same path as in the robust regime. The sum is the result that matters. Total persuasion spending rises from 4,228 in the all-human market to a peak of **7,119** at  $\alpha = 0.3$ , which is **1.68 times** the pre-agent advertising budget, and settles at 5,102 in the all-agent market, still **1.21 times** the pre-agent level. Sellers spend more to persuade agents than they ever spent to persuade people, in a market where margins have fallen by three quarters.

This is the mechanism of Proposition 2 operating in equilibrium: the sharper the agent, the more concentrated its demand, and the higher the marginal return to being the seller it selects. The migration is not a substitution at constant cost — it is an escalation.

## 7.3 What persuadable agents cost: rent and concentration, not misallocation

Here the corrected results change the story, and the change is worth stating precisely because an earlier version of this paper claimed something stronger.

Persuadable agents do **not** meaningfully degrade which sellers win. At  $\alpha = 1$  the top-three true-value sellers take 84.5% of sales under robust agents and 85.9% under manipulable ones. A well-informed agent that can be persuaded still buys from good sellers, because agent-directed spending shifts choice at the margin rather than overturning large quality-adjusted price differences.

What persuadable agents cost is captured on three other measures.

**Rent survives.** Margin at  $\alpha = 1$  is 9.9% under manipulable agents against 4.0% under robust ones — sellers retain roughly two and a half times the margin. The attention rent does not disappear; it is re-earned through a different instrument.

**Consumers capture far less.** Consumer net value per unit at  $\alpha = 1$  is  $+0.23$  under manipulable agents against  $+1.09$  under robust ones, a reduction of about 80%. The market allocates to good sellers and then hands most of the resulting surplus to those sellers.

**The market concentrates.** HHI at  $\alpha = 1$  reaches 7,424 under manipulable agents against 3,102 under robust ones — 2.4 times more concentrated, and near-monopoly in absolute terms. Persuadability rewards scale in agent-directed spending, and scale compounds.

Price dispersion also nearly vanishes under manipulable agents (CV 0.050 against 0.098), which is worth flagging because *low dispersion is often read as evidence of competition*. Here it coexists with the most concentrated and least consumer-favorable configuration in the parameter space.

So the harm from persuadable buyer agents, on these results, is distributional and structural rather than allocative: the right sellers win, at prices that keep the rent with them, in a market that collapses toward a single supplier.

## 7.4 The persuadability–adoption plane

The two-regime comparison above fixes  $\gamma$  at 0 or 1.5. Figure 2 sweeps the plane, with  $\gamma \in 0, 0.25, 0.5, 0.75, 1, 1.5, 2, 3$  against the same adoption path, 4 seeds per cell. Three features of the surface matter more than the two-point contrast.

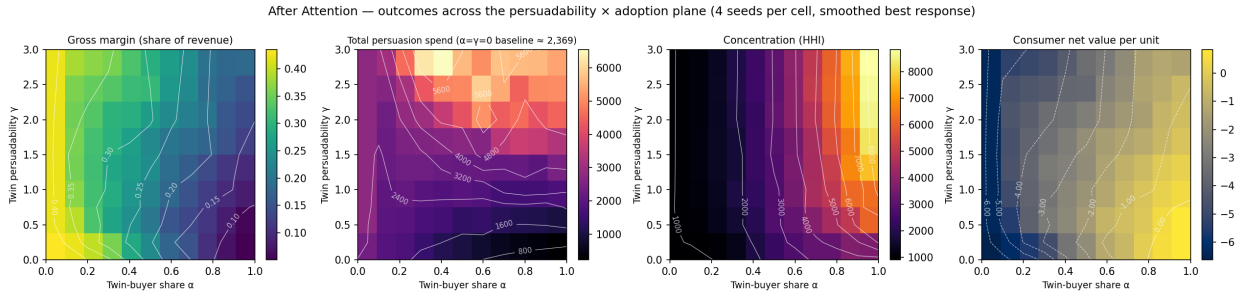


Figure 2. Outcomes across the persuadability  $\times$  adoption plane: gross margin, total persuasion spend, concentration and consumer net value per unit, for  $\gamma$  from 0 to 3 against twin share  $\alpha$  (4 seeds per cell). White contours are iso-outcome lines.

**Rent at full adoption is monotone and graded in  $\gamma$ .** Gross margin at  $\alpha = 1$  rises steadily with persuadability: 5% at  $\gamma = 0$ , 6% at 0.25, 8% at 0.5, 10% at 0.75, 12% at 1, 14% at 1.5, 15% at 2, and 18% at 3. There is no threshold here — every increment of persuadability buys the seller side margin it would otherwise lose.

**Concentration responds to very small persuadability.** HHI at  $\alpha = 1$  is 3,617 under robust agents and 6,089 at  $\gamma = 0.25$  — mild persuadability, the kind that would be hard to detect in an audit, nearly doubles concentration. Beyond that the surface flattens: 8,273 at  $\gamma = 1$  and 8,834 at  $\gamma = 3$ . Most of the structural damage is done by the first small departure from robustness.

**Consumer net value crosses zero.** At full adoption, consumer net value per unit is \$+0.83\$ under robust agents, \$+0.44\$ at  $\gamma = 0.5$ , \$+0.18\$ at  $\gamma = 0.75$ , and \$-0.24\$ at  $\gamma = 1$ , continuing to fall thereafter. Between  $\gamma = 0.75$  and  $\gamma = 1$  the fully-agentic market stops delivering consumers more value per unit than the human market it replaced. The level of this series is not meaningful (Appendix C), but the crossing point is: there is a persuadability above which agent adoption is not an improvement for buyers on this measure, and it is not an extreme value of the parameter.

Total persuasion spending shows the same escalation as §7.2 across the plane, peaking at 2.8 times the  $\alpha = \gamma = 0$  benchmark in the  $\gamma = 3$  row. The peak moves left as  $\gamma$  rises — more persuadable agents pull the arms race earlier in the adoption path.

## 7.5 Structural sensitivity

Table 2 perturbs the four structural parameters one at a time and reports the outcomes that carry the paper's claims. The question is not whether magnitudes move — they do — but whether the sign and rough size of the  $\gamma$  effect survive.

**Table 2. Structural sensitivity (values at  $\alpha = 1$ ; 3 seeds, \$M = 1\{, \}500\$)**

| Perturbation                           | Margin, $\gamma=0$ | Margin, $\gamma=1.5$ | HHI, $\gamma=1.5$ | Peak persuasion, $\gamma=1.5$ |
|----------------------------------------|--------------------|----------------------|-------------------|-------------------------------|
| $\theta_q = 8$ (quality matters less)  | 0.060              | 0.227                | 8,782             | 4,833                         |
| $\theta_q = 16$ (quality matters more) | 0.061              | 0.157                | 8,668             | 4,713                         |
| $\lambda = 0.7$ (less price-sensitive) | 0.067              | 0.182                | 7,874             | 5,983                         |
| $\lambda = 1.4$ (more price-sensitive) | 0.007              | 0.228                | 8,989             | 3,637                         |
| $\beta = 0.6$ (weak ad persuasion)     | 0.048              | 0.131                | 8,792             | 4,464                         |
| $\beta = 2.0$ (strong ad persuasion)   | 0.048              | 0.131                | 8,792             | 4,868                         |
| $\mu_t = 2$ (blunter agents)           | 0.058              | 0.143                | 7,406             | 4,340                         |
| $\mu_t = 5$ (sharper agents)           | 0.036              | 0.140                | 9,177             | 4,908                         |

In every configuration, persuadable agents leave sellers between 2.2 and 32 times the margin that robust agents leave them, and concentration at full adoption lands between 7,400 and 9,200. The most price-sensitive configuration ( $\lambda = 1.4$ ) produces the starkest contrast, because robust agents drive margin almost to zero there while persuadable ones do not.

Two internal checks are worth noting. The  $\beta$  rows are identical at  $\alpha = 1$ , as they must be: with no humans left, the human persuasion weight cannot matter, and the code reproduces that exactly. And sharper agents ( $\mu_t = 5$ ) raise concentration while lowering robust-agent margin — both directions predicted by Propositions 1 and 2, where  $\mu_t$  enters the collapse of margins and the return to persuading agents through the same channel.

## 7.6 Three variants: elastic demand, mixed agent quality, and conflicted agents

The results so far hold the buyer population and category demand fixed. Table 3 relaxes each in turn. All three variants run on the same reduced configuration as §7.4 (\$M = 1\{, \}500\$, 12-point markup grid, 5 seeds), and should be compared against that configuration's baseline rather than against Table 1.

**Table 3. Variants at full agent adoption ( $\alpha = 1$ )**

|                             | Baseline   |              | Mixed quality |              | Conflicted agents |              |
|-----------------------------|------------|--------------|---------------|--------------|-------------------|--------------|
|                             | $\gamma=0$ | $\gamma=1.5$ | $\gamma=0$    | $\gamma=1.5$ | $\gamma=0$        | $\gamma=1.5$ |
| Gross margin                | 0.05       | 0.14         | 0.05          | 0.10         | 0.05              | 0.12         |
| Concentration (HHI)         | 3,617      | 8,567        | 3,722         | <b>4,437</b> | 3,918             | 4,375        |
| Top-3 true-value share      | 0.88       | 0.68         | 0.88          | 0.70         | 0.88              | <b>0.60</b>  |
| Consumer net value per unit | +0.83      | −0.46        | +0.89         | <b>+0.03</b> | +0.62             | <b>−0.67</b> |
| Total persuasion spend      | 224        | 4,497        | 256           | 2,813        | 613               | 3,295        |

**Elastic category demand.** When the outside option strengthens with the market price level, the transition costs the category volume — but only under robust agents. Units sold fall from 1,492 at  $\alpha = 0$  to 951 at  $\alpha = 0.4$ , recovering to 1,264 at full adoption as prices fall. Under persuadable agents volume never contracts at all (1,490 to 1,500 across the whole path), because agent-directed spend enters twin utility and props up the purchase decision. Persuasion buys back the volume that price competition would otherwise cost the category. This is the agentic-commerce counterpart of the long-standing finding that advertising expands category demand, and it is worth flagging for anyone reading category growth as evidence that agent

adoption is going well.

**Mixed agent quality.** When half the twin population is robust and half is persuadable, the robust half does substantial work. Concentration at full adoption falls from 8,567 to 4,437 — roughly half the structural damage removed by half the population — and consumer net value per unit returns to approximately break-even (+0.03 against −0.46). Seller margin falls by less, from 0.14 to 0.10. Robust agents therefore generate a positive externality for buyers whose agents are not robust, and they do it more effectively on market structure than on price. A policy that raises the share of resistant agents does not need to reach every buyer to matter.

**Conflicted agents.** The third variant replaces the robust half of that mixture with a platform-owned agent that weights a per-sale commission offered by the seller. The commission is a transfer out of the seller's margin rather than a fixed cost, which is what distinguishes an agent with divided loyalty from one that is merely persuadable. Against the mixed-quality variant — the same 50/50 population structure — conflicted agents cut the top-three true-value share from 0.70 to **0.60** and consumer net value per unit from +0.03 to **−0.67**, the worst figure anywhere in our results. Sellers pay about 520 in commissions at full adoption.

This is the misallocation that §7.3 declined to attribute to persuadability. Persuadable agents mostly relocate rent; conflicted agents actually reroute demand, because the seller can pay for the choice directly rather than having to move it through content the agent might discount. If the deployed configuration is agents owned by parties with a commercial interest in the transaction — which the rails of §3 make easy to build and do nothing to constrain — the harm is larger and different in kind from the persuadability case.

## 7.7 Integrity dominates structure (P3)

The two regimes differ in exactly one parameter, yet at high agent adoption they produce markets that differ by a factor of 2.5 in margin (9.9% against 4.0%), a factor of 2.4 in concentration (HHI 7,424 against 3,102), and a factor of nearly 5 in consumer net value per unit (+0.23 against +1.09). Nothing about the number of sellers, their costs, or buyers' underlying preferences differs between the two columns. Who captures the gains from agent-mediated commerce is, in this model, a property of the buyer's agent rather than of the market's structure.

That has an uncomfortable corollary for policy. Competition policy, advertising disclosure, and platform regulation all address the market's supply side; none of them touches  $\gamma$ . If the relevant parameter is how resistant buyer agents are to seller persuasion, then the effective regulatory object is the agent's loyalty to its principal — which is measurable in principle (a susceptibility test on matched offers) and certifiable in principle (a published resistance score, attested rather than self-declared). We regard this as the main policy implication of the paper, and note that it is being approached legislatively from a different direction: the AI AGENT Act of 2026 (S. 5051, introduced 21 July 2026) would guarantee agent access to platforms and create an FTC registry of trusted agents, but access and registration are not resistance.

## 8. What replaces advertising

If robust agents make consumer advertising unsellable in a category, sellers still need to be chosen. Four instruments plausibly absorb the spend, and they differ sharply in whether they route demand to value.

1. **Verifiable claims.** Signed, machine-checkable attributes (provenance, delivery commitments, test results) that an agent can verify instead of reading persuasive prose. This is the value-routing substitute: it makes truth cheaper to transmit than persuasion. As §3 shows, this is precisely the layer the current rails leave unbuilt: they sign the transaction envelope and the payment authorization, not the claim. Lineage-gated attestation architectures developed for enterprise data governance (Sangaraju and Vissa

2026, preprint) offer a technical template — derivation-gated access control that binds a claim's validity to its verified provenance chain rather than to the claimant's own assertion — adaptable here to bind a product claim's trustworthiness to an attested source instead of the seller's word.

2. **Agent-readable reputation.** Structured outcome histories — return rates, delivery performance, warranty claims — auditable rather than curated. Agents reward this only if it is attested; otherwise it degrades into the same optimization target as description text.
3. **Agent-to-agent negotiation.** Price and terms set through protocol rather than posted. This shifts rent toward whoever has better information about the counterparty, which is a different distributional question from attention rent.
4. **Agent-directed optimization.** The  $G$  of our model: content and structure engineered for agent consumption. This is the instrument that Proposition 2 predicts will be funded. On our results it does not chiefly reroute demand away from good sellers; it raises what those sellers can charge and concentrates the market around whoever spends most. Generative engine optimization (Aggarwal et al. 2024) is its existing commercial form.

Which of these dominates is not a technological question but a question of what agents verify. An agent that discounts unverified claims makes instrument 1 profitable; an agent that treats all text alike makes instrument 4 profitable. This is the design margin on which the market's efficiency turns.

## 9. Implications

**Brands.** The knee in §7.1 means early agent adoption is not a signal of safety. Note also that the defensive value of persuading agents is real in our results: manipulable-agent markets leave sellers roughly two and a half times the margin. What is privately rational here is collectively costly, which is the usual shape of an arms race. The defensible position under robust agents is genuine quality-adjusted value, which our model rewards heavily: the best-value sellers roughly double their share of sales as agents take over. The brand premium that survives is the part backed by attributes an agent can verify.

**Retailers and platforms.** Under robust agents, the margin available from obfuscation (Ellison and Ellison 2009) goes to zero. Under persuadable agents, whoever controls the surface an agent reads captures the migrated persuasion spend — which is a reason to expect platforms to prefer persuadable agents, and a reason to watch rail consolidation (AP2, ACP, the Universal Commerce Protocol) as a market-structure question rather than a plumbing one. The rails as specified (§3) do not constrain this: whoever defines the surface an agent reads captures the migrated spend, and no published protocol requires that surface to separate attested fact from promotion.

**Advertising platforms.** The \$1.3 trillion figure is not a single pool that either survives or disappears; it is a pool that migrates at a rate set by  $\gamma$  and  $\alpha$ . Our manipulable-regime result — total persuasion spending peaking at 1.68 times the pre-agent level — implies the intermediate transition is a growth period for agent-directed persuasion services even as consumer advertising declines.

**Regulators.** The object to regulate is agent resistance, not agent access. A registry of agents that lists identity without attesting susceptibility does not constrain the mechanism in P2. The same holds for the standards: mandating protocol adoption raises  $\alpha$  and leaves  $\gamma$  untouched, which by our results is the combination that produces the worst outcomes.

**Platform-owned agents.** The conflicted-agent variant of §7.6 is the configuration that produces genuine misallocation, and it is also the one the current rails make easiest to deploy: a commission paid to the agent's operator needs no attested claim, no registry and no change to any published protocol. Whatever separates

an agent's revenue from the seller's choice — fee-for-service pricing, disclosure of commissions to the principal, or an enforced duty of loyalty — matters more for allocation than agent robustness does.

**Agent providers.** P3 is a market-making opportunity: a provider that can *prove* resistance to seller persuasion is selling the parameter that determines whether the market its users buy in allocates well. That proof has to be adversarial and published, not asserted.

## 10. Limitations and next steps

We are explicit about the gap between what we have shown and what we would need to show.

1.  **$\gamma$  is stylized, not measured.** The single most valuable extension is an empirical estimate: run frontier shopping agents on matched product listings that differ only in agent-directed optimization, and estimate the choice-share shift per unit of persuasion. That converts P2 from an assumption into a prediction. The experimental designs in Allouah et al. (2026) and Salvi et al. (2026) are the natural templates.
2. **Equilibrium selection.** Proposition 3 rules out a pure-strategy equilibrium in the interior, so the reported outcomes are averages over the stationary distribution of a smoothed dynamic rather than equilibrium comparative statics. A mixed-strategy characterization of the interior — rather than a numerical approximation to it — remains open.
3. **Stylized parameters.** Magnitudes are illustrative. The directional results (flat-then-cliff margins, migration of persuasion spend, divergence by  $\gamma$ ) are robust across seeds; specific figures such as "1.68 times" are not estimates of any real market.
4. **Single category.** All results are within one product category. The elastic-demand variant of §7.6 lets the category contract, but cross-category substitution — buyers moving spend elsewhere as an agent-mediated category changes — is not modeled.
5. **Two-point agent quality.** The mixed-quality variant of §7.6 uses a 50/50 split of robust and persuadable agents. A full distribution over  $\gamma$ , and endogenous choice of agent by buyers who observe something about agent quality, are both open.
6. **Conflicted agents are modeled simply.** The commission in §7.6 is a flat per-sale rate chosen from a small grid, with no bargaining, no disclosure regime, and no reputational consequence for the agent's operator. The result — that conflicted agents misallocate where persuadable ones do not — is the variant's clearest finding and the one most in need of a richer treatment.
7. **Welfare in the mixed population.** We have not examined whether human buyers are made worse off as agents take the good deals; the model can address this and the result ("a twin divide") would matter for policy.
8. **The propositions are partial-equilibrium.** Proposition 1 and Proposition 2 hold rivals' actions fixed (Assumption 3). A closed-form characterization of  $\alpha^*$  with endogenous rival prices, and of equilibrium agent-directed spending rather than the condition for entry into it, are both open.

## 11. Conclusion

The expectation that purchasing agents will erode the margin that comes from human inattention is correct in direction, but the path is not smooth and the destination depends on something the market is not currently building.

Attention rent does not decline gradually. It survives intact through early agent adoption and then falls over a narrow band, because a well-informed agent's demand confronts sellers with a discrete choice rather than a continuous trade-off. In the transition region the mixed population supports no pure-strategy equilibrium at

all — sellers are caught between harvesting buyers who do not compare and undercutting for buyers who do, which is the structure Varian identified in 1980 and which our numerics reproduce exactly.

The persuasion spending that attention rent funded does not disappear either. If agents can be moved by how a listing is written, the spending migrates to moving agents, and in our simulations it peaks at 1.68 times its pre-agent level. What that migration buys is not a market that picks the wrong sellers — good sellers still win about 85% of sales whether or not agents are persuadable. What it buys is a market where those sellers keep two and a half times the margin, consumers capture roughly 80% less value per unit, and concentration more than doubles. The rent is not dissolved by making buyers smarter; it is relocated, and it flows to whoever the agents can be induced to choose.

What separates the two outcomes is not competition, search cost, or regulation of sellers. It is whether the buyer's agent can be persuaded. The protocols standardized so far secure the payment and leave the claim unexamined, which is precisely the configuration in which the second outcome obtains. That makes the integrity of buyer agents — verifiable, adversarially tested, and published rather than asserted — the central economic institution of an agent-mediated consumer market.

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## Appendix A. Proofs

Throughout,  $M$  is the number of buyers,  $\alpha$  the twin share, and seller  $j$  has cost  $c_j$  and price  $p_j$ . Assumptions 1–3 are as stated in §5.

**Proof of Proposition 1.** By Assumption 1 the seller compares two candidates. At  $p_H$  it earns per-human profit  $\Pi_H$  from the  $(1-\alpha)M$  humans and, since  $p_H > p_T$  and  $p_T$  is by construction the highest price winning twin demand, Assumption 2 gives it approximately none of the  $\alpha M$  twins. Its profit is therefore

$$V_H(\alpha) = (1-\alpha) M \Pi_H.$$

At  $p_T$  it earns  $\pi^h_T$  per human and  $\pi^t_T$  per twin:

$$V_T(\alpha) = (1-\alpha) M \pi^h_T + \alpha M \pi^t_T.$$

Both are affine in  $\alpha$ . Then  $V_H(\alpha) \geq V_T(\alpha)$  iff  $(1-\alpha)(\Pi_H - \pi^h_T) \geq \alpha \pi^t_T$ , i.e. iff  $(1-\alpha)\Delta \geq \alpha \pi^t_T$  with  $\Delta = \Pi_H - \pi^h_T$ . Rearranging,  $\Delta \geq \alpha(\Delta + \pi^t_T)$ , so

$$\alpha \leq \Delta / (\Delta + \pi^t_T) = \alpha^*_j.$$

Since  $\Delta > 0$  and  $\pi^t_T > 0$ , both numerator and denominator are positive and the numerator is strictly smaller, so  $\alpha^*_j \in (0,1)$ . Differentiating,

$$(\partial \alpha^*_j) / (\partial \pi^t_T) = -(\Delta) / ((\Delta + \pi^t_T)^2) < 0, (\partial \alpha^*_j) / (\partial \Delta) = (\pi^t_T) / ((\Delta + \pi^t_T)^2) > 0. \blacksquare$$

**Proof of Corollary 1.**  $V_H$  and  $V_T$  are affine in  $\alpha$  with  $V_H$  decreasing in  $\alpha$  at rate  $M\Pi_H$  and  $V_T$  changing at rate  $M(\pi^t_T - \pi^h_T)$ . They cross exactly once, at  $\alpha^*_j$ , by Proposition 1. The argmax therefore switches from  $p_H$  to  $p_T$  at that point and the realized price, and hence the realized margin, is a step function of  $\alpha$  with a single jump. Aggregate margin is the sales-weighted average of individual margins; a finite sum of step functions with distinct jump points is a step function whose increments are located at the individual thresholds, which for a dispersed distribution of  $\alpha^*_j$  appears as a steep but continuous knee.  $\blacksquare$

**Proof of Proposition 2.** Under the logit specification of §4, a twin chooses  $j$  with probability

$$s^t_j = (\exp(\mu_t(V_0 + \theta_q q_j - \lambda p_j + \gamma \log(1+G_j)))) / (\sum_k \exp(\mu_t(V_0 + \theta_q q_k - \lambda p_k + \gamma \log(1+G_k)))).$$

Holding rivals fixed (Assumption 3), the standard logit derivative gives

$$(\partial s^t_j) / (\partial G_j) = \mu_t \gamma (s^t_j(1-s^t_j)) / (1+G_j).$$

Seller  $j$ 's profit contribution from the twin segment net of agent-directed spend is  $\alpha M (p_j - c_j) s^t_j - G_j$ , so

$$(\partial) / (\partial G_j) [\alpha M (p_j - c_j) s^t_j - G_j] = \alpha M (p_j - c_j) \mu_t \gamma (s^t_j(1-s^t_j)) / (1+G_j) - 1.$$

Evaluating at  $G_j = 0$ , spending is strictly profitable at the margin iff

$$\alpha M (p_j - c_j) \mu_t \gamma s^t_j(1-s^t_j) > 1,$$

which rearranges to  $\gamma > \gamma^*_j$  with  $\gamma^*_j$  as stated. The right-hand side is a positive constant divided by  $\alpha M (p_j - c_j) \mu_t s^t_j(1-s^t_j)$ , so  $\gamma^*_j$  is strictly decreasing in each of  $\alpha$ ,  $(p_j - c_j)$  and  $\mu_t$ , holding the others

and  $s^{\wedge}t_j$  fixed. ■

**Proof of Proposition 3.** Write  $\Pi(p \text{ win}) = (p-c)(H/2 + T)$  for the profit of a seller holding both its captive humans and the whole twin segment, and  $\Pi(p \text{ lose}) = (p-c)H/2$  for a seller holding only its captive humans. We show every profile admits a profitable deviation.

*Case A:*  $p_1 = p_2 = p$ . Each seller earns  $\$(p-c)(H/2 + T/2)\$$ . Undercutting to  $p - \delta$  yields  $(p-\delta-c)(H/2+T)$ , so the gain is

$$(p-c)(T)/(2) - \delta((H)/(2)+T).$$

This is positive whenever  $(p-c)T > 2\delta(H/2+T)$ , which holds at  $p = v_H$  by (i). If instead  $p$  is low enough that undercutting does not pay, then  $(p-c) \leq 2\delta(H/2+T)/T$ , and deviating up to  $v_H$  — abandoning the twin segment and harvesting captive humans — yields  $(v_H-c)H/2$  against a current payoff of at most  $2\delta(H/2+T)(H/2+T/2)/T$ , so the deviation is profitable by (ii). Every symmetric profile therefore admits a profitable deviation.

*Case B:*  $p_1 < p_2$ . Seller 2 sells only to its captive humans, earning  $(p_2-c)H/2$ , which is strictly increasing in  $p_2$ ; so unless  $p_2 = v_H$ , raising is profitable. Given  $p_2 = v_H$ , seller 1 holds the twin segment at any  $p_1 < v_H$  and its profit  $(p_1-c)(H/2+T)$  is strictly increasing in  $p_1$ ; so unless  $p_1 = v_H - \delta$ , raising is profitable. The only surviving candidate is  $(v_H - \delta, v_H)$ . At that profile seller 2 earns  $(v_H-c)H/2$ ; deviating to  $v_H - 2\delta$  captures the twin segment and yields  $(v_H-2\delta-c)(H/2+T)$ , a gain of

$$(v_H-c)T - 2\delta((H)/(2)+T) > 0$$

by (i). So this profile is not an equilibrium either, and no pure-strategy equilibrium exists. ■

**Proof of Corollary 2.** Substitute  $T = \alpha M$  and  $H = (1-\alpha)M$ . Condition (i) becomes  $\alpha(v_H - c) > \delta(1+\alpha)$ , which defines a lower bound  $\alpha = \delta/(v_H - c - \delta)$ , decreasing in  $(v_H-c)/\delta$  and tending to  $0$  as  $\delta \rightarrow 0$ . Condition (ii) becomes an upper bound  $\alpha$ : its left side is proportional to  $(1-\alpha)$  and vanishes as  $\alpha \rightarrow 1$ , while its right side is bounded away from zero for fixed  $\delta$ , so (ii) fails near  $\alpha = 1$  and holds below some  $\alpha < 1$ ; as  $\delta \rightarrow 0$  the right side vanishes and  $\alpha \rightarrow 1$ . Hence the non-existence interval  $(\alpha, \alpha)$  is non-empty when  $\delta$  is small relative to  $v_H - c$ , and expands to  $(0,1)$  in the limit. At  $\alpha = 0$  there is no twin segment, so Case A's undercutting gain is  $-\delta H/2 < 0$  and the symmetric profile at  $v_H$  is an equilibrium; at  $\alpha = 1$  there are no captive humans, condition (ii) fails, and the grid's lowest profitable price is an equilibrium. ■

**Remark.** The derivative is taken at  $G_j = 0$  and treats  $s^{\wedge}t_j$  parametrically. Because  $s^{\wedge}t_j$  itself responds to rivals' spending, the condition characterizes entry into agent-directed spending, not the equilibrium level of it. Equilibrium levels are obtained numerically in §7.

## Appendix B. Simulation implementation and parameters

| Parameter  | Value                | Meaning                        |
|------------|----------------------|--------------------------------|
| $N$        | 12                   | sellers                        |
| $M$        | 3,000                | buyers                         |
| $q_j$      | $\$U(0,1)\$, sorted$ | true quality                   |
| $c_j$      | $5 + 10q_j$          | marginal cost                  |
| $\theta_q$ | 12                   | value of quality               |
| $\lambda$  | 1                    | price sensitivity (both types) |
| $V_0$      | 5                    | category baseline value        |

| Parameter        | Value                | Meaning                                               |
|------------------|----------------------|-------------------------------------------------------|
| $\beta$          | 1.2                  | human persuasion weight                               |
| $\sigma\epsilon$ | 1.0                  | human quality-perception noise                        |
| $\mu_t$          | 3.0                  | twin logit scale (choice sharpness)                   |
| $a_0, a_1$       | -1.5, 0.9            | human consideration function                          |
| $\gamma$         | 0 or 1.5             | twin persuadability                                   |
| markup grid      | 0.01–1.7 (16 points) | seller price choice                                   |
| $A, G$ grids     | 0–1,500 (8 points)   | persuasion spend choices                              |
| iterations       | 40, last 10 averaged | best-response dynamic                                 |
| seeds            | 20                   | per $(\alpha, \gamma)$ cell                           |
| $\tau$           | 0.15                 | smoothing temperature (relative to profit dispersion) |
| $\iota$          | 0.35                 | probability a seller skips a revision                 |

**Smoothed best-response dynamic.** Sellers are visited in a randomized order each iteration and skip the revision with probability  $\iota$ . A revising seller evaluates every combination of markup, advertising level and agent-spend level on the grids, holding rivals fixed, and draws its action from a logit distribution over the resulting profits with temperature  $\tau$  scaled by the standard deviation of those profits. Exact best response is the  $\tau \rightarrow 0$  limit; we use  $\tau = 0.15$  because Proposition 3 rules out a pure-strategy equilibrium in the interior, so the exact dynamic has no fixed point to find there. Common random numbers are drawn once per run — consideration draws, quality-perception errors, and extreme-value taste shocks for both buyer types — and reused across all candidate actions, so that differences in evaluated profit reflect the action rather than sampling noise. The dynamic runs 40 iterations, or stops early if a full sweep produces no change, which is a pure-strategy fixed point.

**Reporting.** Outcomes are computed at each of the final 10 of 40 iterations and averaged, then averaged again over 20 seeds.

**Reproduction.** `python3 sim.py 5` writes `results.json` (110 runs); `python3 plot.py` produces `fig_outcomes.png` and the summary table. Every figure in §7 is a mean over the 5 seeds of the final-10-iteration averages recorded in `results.json`.

## Appendix C. Robustness and known fragilities

**Non-convergence is predicted, and observed exactly where predicted.** Under exact best response on the earlier configuration, 12 of 110 runs reached a pure-strategy fixed point. All 12 sit at  $\alpha = 0$  or  $\alpha = 1$ ; none of the 90 interior runs converged. This is Proposition 3 and Corollary 2 in the data: the interior has no pure-strategy equilibrium, the boundaries do. The smoothed dynamic used for the results in §7 does not require convergence — it averages over a stationary distribution — so the reported numbers no longer depend on where a cycle is cut.

**What the smoothing costs.** A revising seller draws from a logit distribution rather than maximizing, so it sometimes funds instruments with no return. The clearest measure of this is agent-directed spend at  $\alpha = 0$ , where there are no agents to persuade: the smoothed dynamic still records about 300 units. Treat roughly 300 as the noise floor for the spend series; differences of that order should not be read as findings. The corresponding figure for advertising at  $\alpha = 1$  is 247, consistent.

**Grid effects, and one artifact we removed.** Prices, advertising and agent spend are chosen from discrete grids. The coarse grid used in an earlier draft (12 markup points) produced a spurious result: in the fully manipulable market, the top-three true-value sellers appeared to take only 4.9% of sales. On a 16-point grid that figure is 85.9%, and the change does not come from the smoothing — under exact best response on the finer grid it is 85.4%. With too few price points, one seller outspending on agent-directed optimization can take the whole twin segment and rivals have no fine enough response available. The two-candidate structure behind Proposition 1 is a real feature of winner-take-most demand; the sharpness of any particular discrete outcome is partly the grid.

**The elastic variant needed calibration, and that is a judgment call.** The outside option's price sensitivity is a free parameter. At scale 0.4 the category collapses entirely under robust agents at full adoption (125 of 1,500 units); at 0.15 it contracts and recovers, which is the behavior we report. We chose 0.15 because it produces contraction without extinction, not because it is estimated from anything. The qualitative finding — that persuadable agents sustain category volume robust agents do not — holds at every scale we tried, but its magnitude is a choice.

**Variant results are exploratory.** The three variants of §7.6 run on the reduced configuration (1,500 buyers, 12-point grid, 5 seeds), which §7.3's grid discussion shows is coarse enough to exaggerate discrete outcomes. Compare them against the same-configuration baseline, which is what Table 3 does, and treat the direction rather than the magnitude as the finding. The top-three value share at  $\gamma = 1.5$ ,  $\alpha = 1$  is 0.68 on this configuration and 0.86 on the finer headline configuration — the variant comparisons are internally consistent, but not directly comparable to Table 1.

**Seed dispersion.** Results are means over 20 seeds; the shaded bands in Figure 1 are  $\pm 1$  standard deviation across seeds. The qualitative pattern — flat-then-falling margins under  $\gamma = 0$ , and under  $\gamma = 1.5$  rising total persuasion spend with higher retained margin and higher concentration — holds in every seed. Point magnitudes do not.

**Scale of outcomes.** Consumer net value per unit is negative at low  $\alpha$  under our parameterization, because  $V_0 + \theta_q q_j$  is small relative to equilibrium prices at high markups. Only the direction and relative movement of that series should be read; its level is an artifact of the value scale.

**Inelastic category demand.** The outside option is rarely chosen, so units sold stay close to  $M$  in the manipulable regime. The model therefore describes how demand is allocated among sellers, not whether the category expands or contracts. Elastic category demand is the most valuable robustness extension.

**Grid effects.** Prices, advertising and agent spend are chosen from coarse grids. This is what makes the two-candidate structure of Proposition 1 visible, and it also exaggerates the discreteness of the switch relative to a continuous-price model. The knee is a real feature of winner-take-most demand; its sharpness is partly a grid artifact.